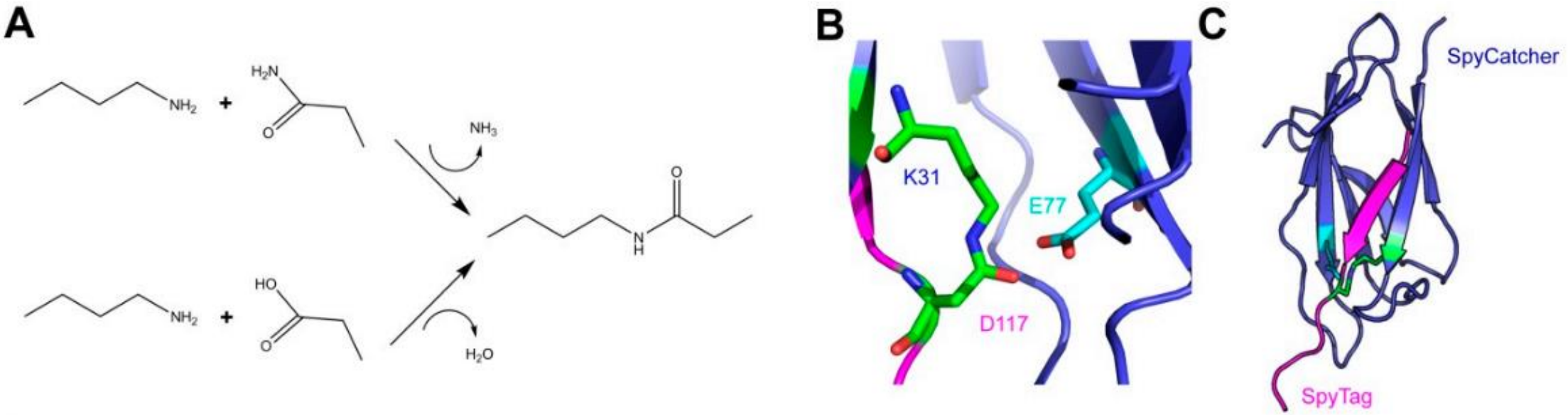
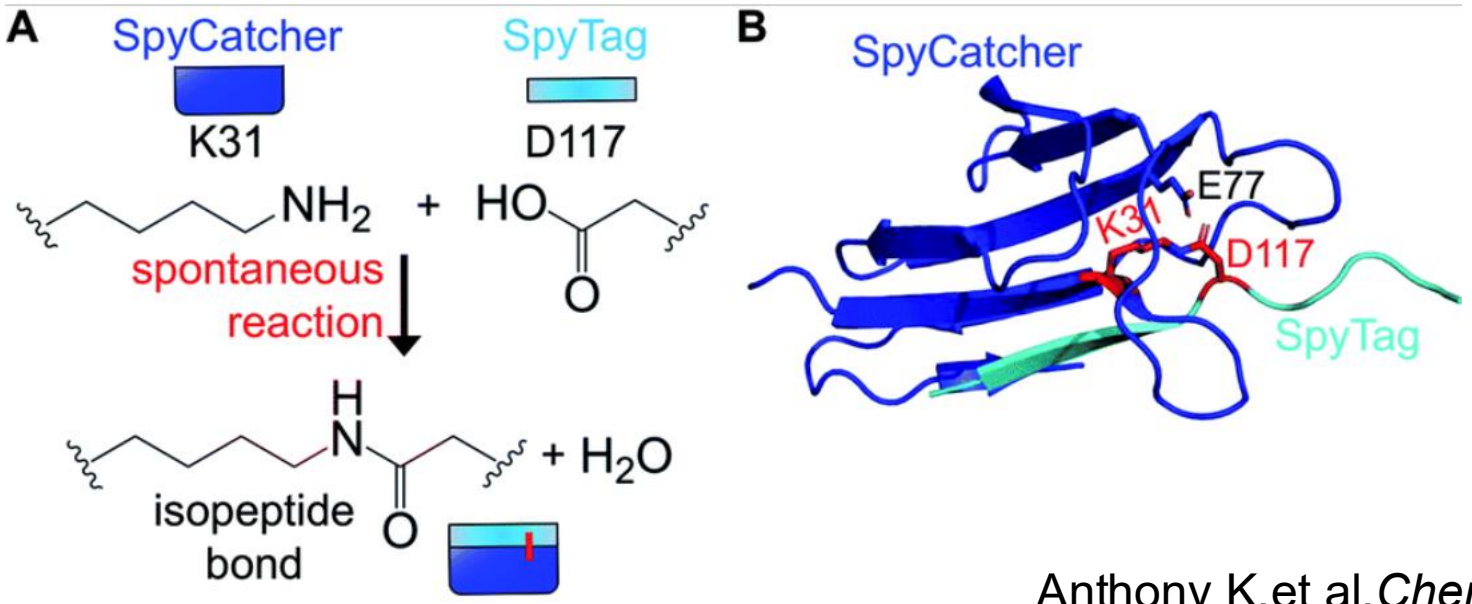
SpyTag-SpyCatcher system



Hatlem D, et al. *Int J Mol Sci*. **2019**

1. The SpyTag-SpyCatcher system, developed from the CnaB2 domain of FbaB in Streptococcus pyogenes, involves a spontaneous internal isopeptide bond between the primary amine of lysine K31 and the side-chain carboxyl of aspartic acid D117.
2. The CnaB2 domain splits into two parts: SpyCatcher, an 113-residue protein with reactive K31, and SpyTag, a 13-residue peptide with reactive D117.

SpyTag-SpyCatcher system

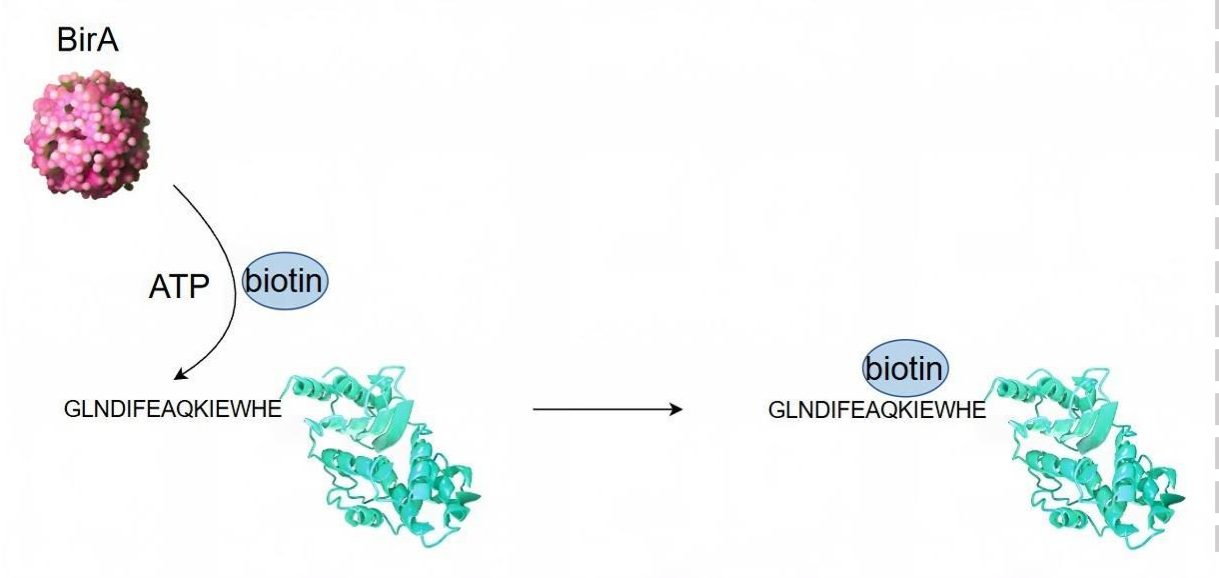


Anthony K,et al.*Chem. Sci.* **2020**

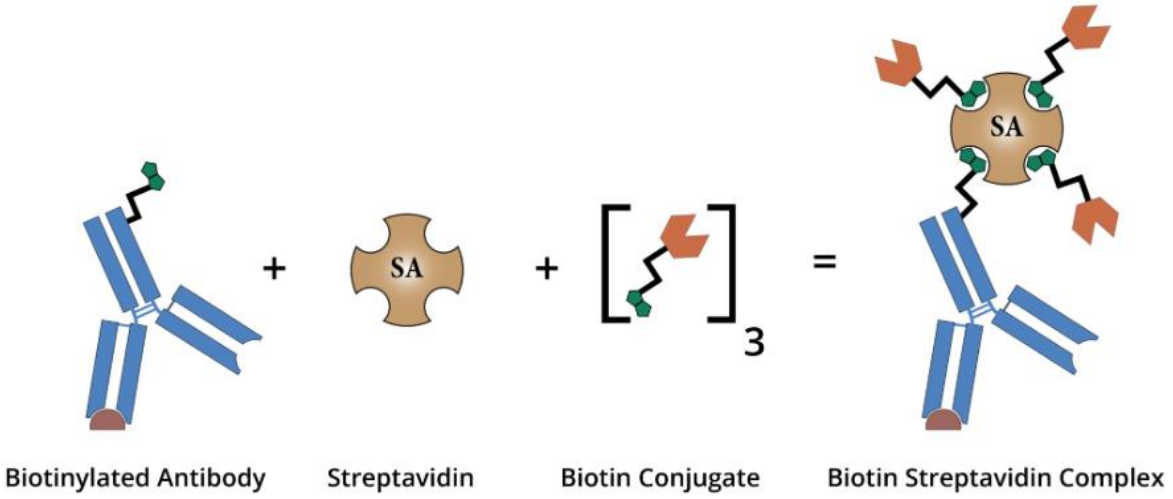
Characteristics of the SpyTag-SpyCatcher system: It enables irreversible coupling with high specificity and up to 99% conversion rate. The interaction occurs over a broad temperature and pH range, with high affinity (0.2 μM). SpyTag and SpyCatcher can be fused to interested proteins for assembly.

Biotin-Streptavidin system

Avi-tag

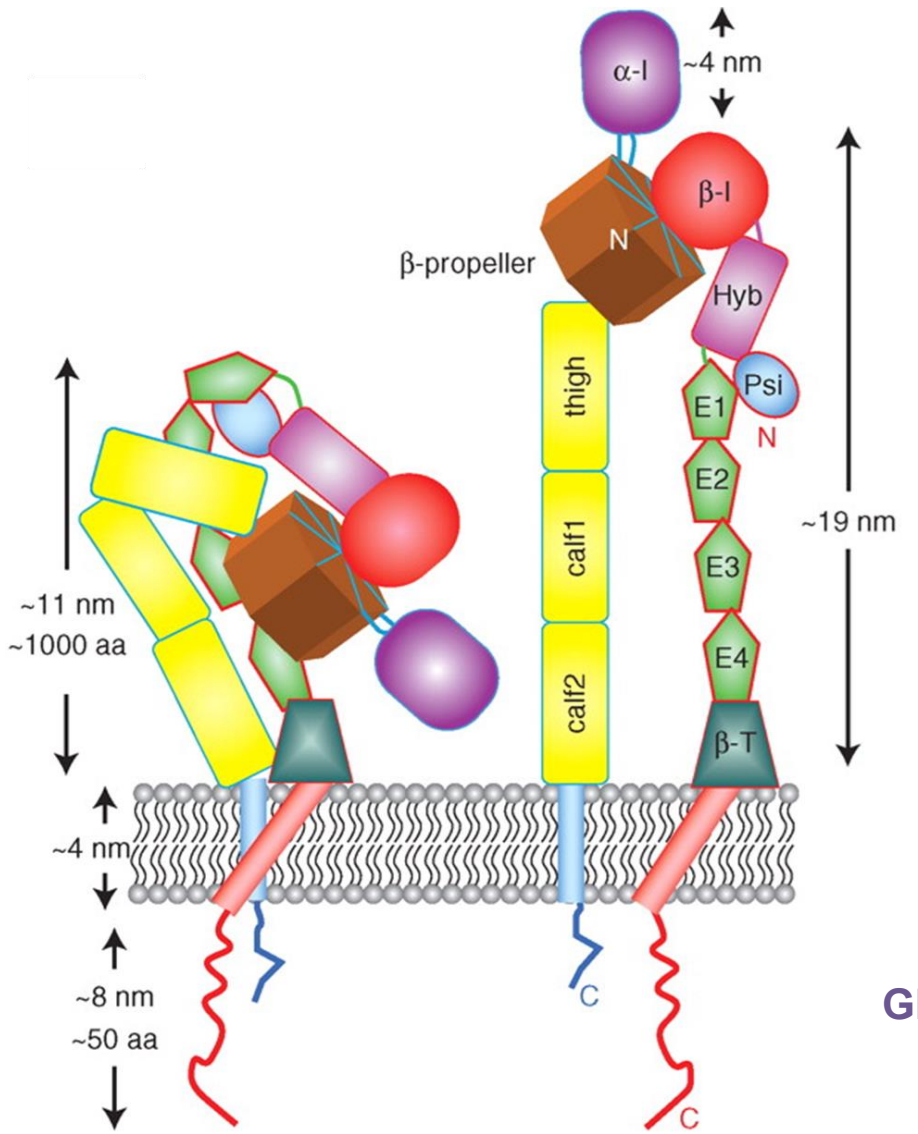


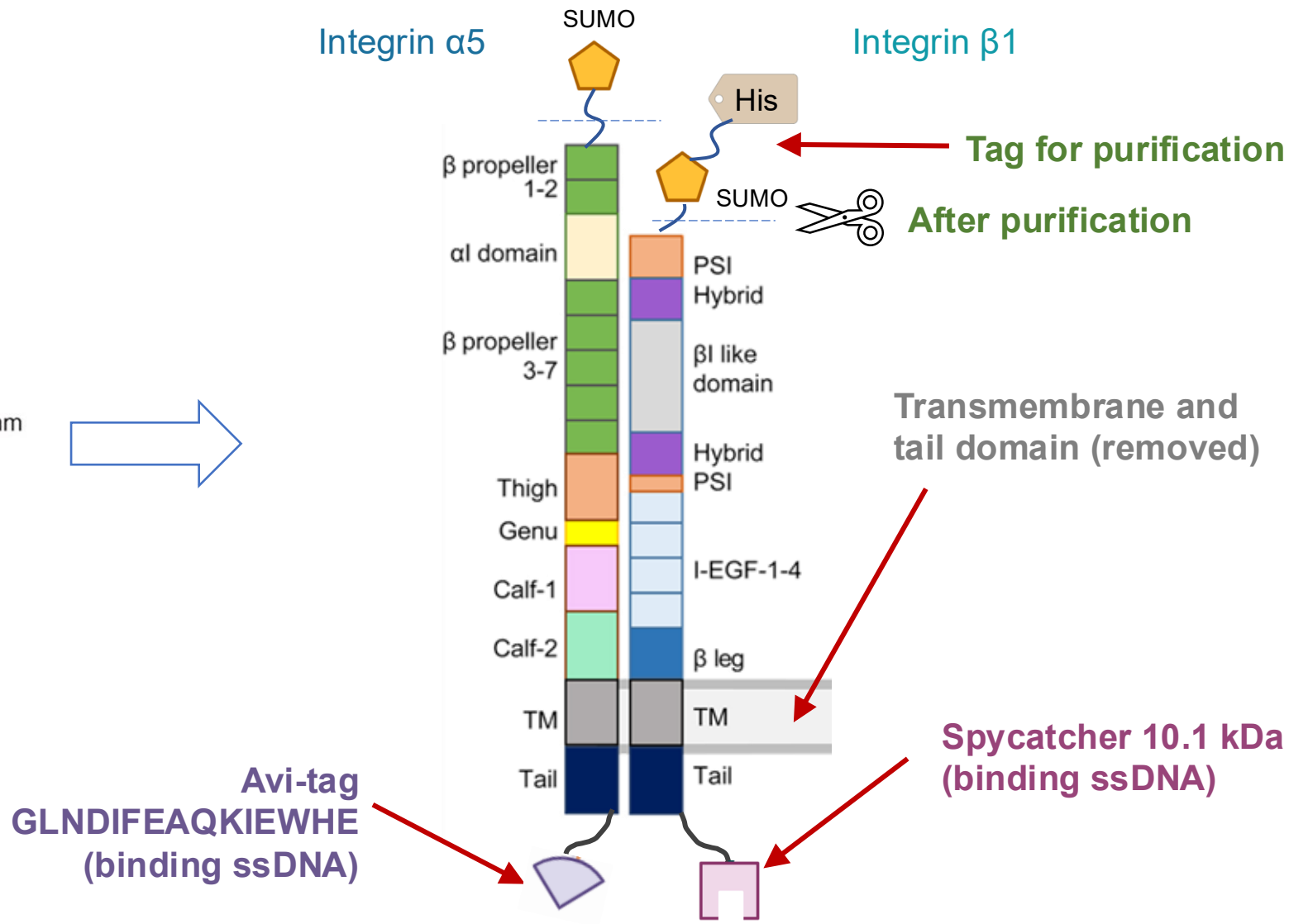
Biotin and Streptavidin



Avi Tag is a short peptide tag composed of 15 amino acid residues (GLNDIFEAQKIEWHE), which contains a biotinylatable lysine residue. Under the action of biotin ligase (BirA), this lysine can specifically and covalently attach a biotin molecule. The biotin then binds to Streptavidin.

Fusion Protein Structure Design of Integrin β1(α5)-Spycatcher(Avi-tag)

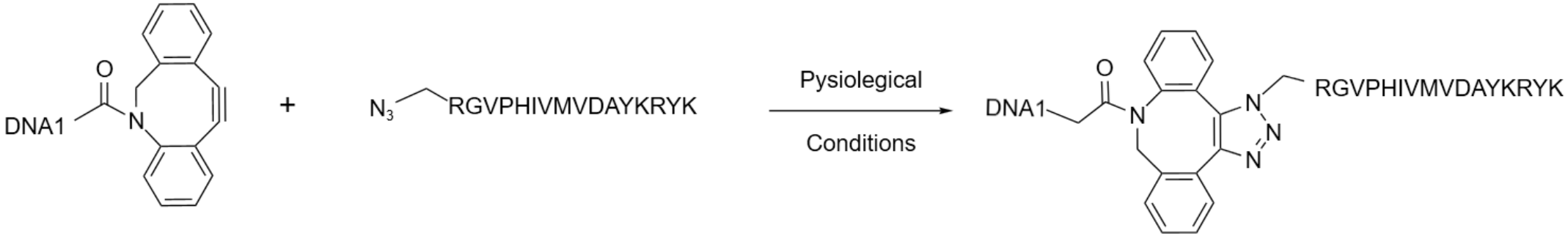




Through Spycatcher/Biotin connect with DNA

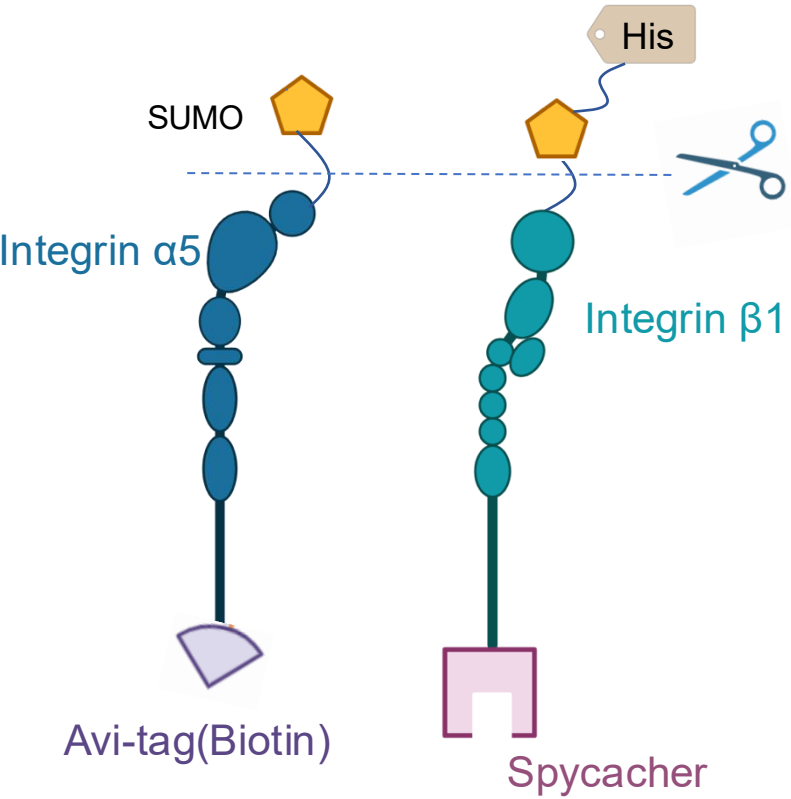
Spytag-DNA1

Spytag AAseq:RGVPHIVMVDAYKRYK



Streptavidin-DNA2

Tag Design



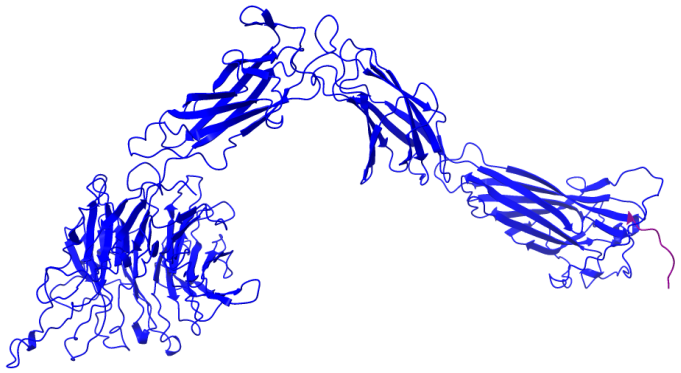
- The SUMO (small ubiquitin-like modifier) tag is a small ubiquitin-like modifier protein that can be used as a fusion tag and molecular chaperone in recombinant protein expression. It not only increases the expression level of the fusion protein but also enhances its solubility and promotes proper folding, while providing protease resistance.
- The Integrin α5 -Avi subunit is constructed in a eukaryotic expression vector containing SUMO tag. After expression, the subunit undergoes biotinylation, enabling purification via streptavidin.
- The Integrin β1 -spycatcher subunit is constructed in a eukaryotic expression vector containing a His-SUMO tag. After expression, the β1 subunit with a His tag can be purified using affinity chromatography.
- Both methods facilitate purification and removal of homodimers or monomers, promoting the formation of the heterodimer. Finally, the SUMO protease is used to cleave the tags, yielding the tag-free heterodimeric protein.

SUMO protease cleavage site

Integrin α5 RQGKEMDSLTFLYDGIEIQADQTPEDLDMEDNDIIEAHREQIGGGSFNLDAEAPAVLSGPPG

Integrin β1 AKRQGKEMDSLTFLYDGIEIQADQTPEDLDMEDNDIIEAHREQIGGGSQTDENRCLKANA

Sequence Design： SUMO-Integrin α5 -Avi AAseq

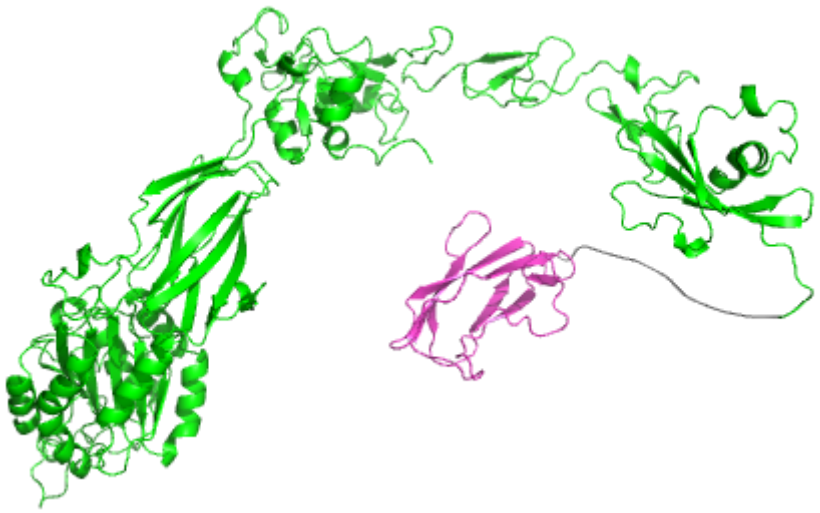


before enzymatic digestion
MW: 117.21 kDa
after enzymatic digestion
MW: 106.05 kDa

MSDSEVNQEAKPEVKPEVKPETHINLKVSDGSSEIFFKIKKTTPLRRLMEAFAKRQGKEMDS
LTFLYDGIEIQADQTPEDLDMEDNDIIEAHREQIGGGSFNLDAEAPAVLSGPPGSFFGFSVEF
YRPGTDGVSVLVGAPKANTSQPGVLQGGAVYLCPWGASPTQCTPIEFDSKGSRLLESSLS
SSEGEEPVEYKSLQWFGATVRAHGSSILACAPLYSWRTEKEPLSDPVGTCYLSTDNFTRILE
YAPCRSDFSWAAGQGYCQGGFSAEFTKTGRVVLGGPGSYFWQGQILSATQEQIAESYYPE
YLINLVQGQLQTRQASSIYDDSYLGYSVAVGEFSGDDTEDFVAGVPKGNLTYGYVTILNGSDI
RSLYNFSGEQMASYFGYAVAATDVNGDGLDDLLVGAPLLMDRTPDGRPQEVGRVYVYLQH
PAGIEPTPTLTLTGHDEFGRFGSSLTPLGDLDQDGYNDVAIGAPFGGETQQGVVFVFPGGP
GGLGSKPSQVLQPLWAASHTPDFFGSALRGGRDLDGNGYPDLIVGSFGVDKAVVYRGRPI
VSASASLTIFPAMFNPEERSCSLEGNPVACINLSFCLNASGKHVADSIGFTVELQLDWQKQK
GGVRRALFLASRQATLTQTLLIQNGAREDCREMKIYLRNESEFRDKLSPIHIALNFSLDPQAP
VDSHGLRPALHYQSKSRIEDKAQILLDCGEDNICVPDLQLEVFGEQNHVYLGDKNALNLTFH
AQNVGEGGAYEAELRVTAPPEAEYSGLVRHPGNFSSLSCDYFAVNQSRLLVCDLGNPMKA
GASLWGGLRFTVPHLRDTKKTIQFDFQILSKNLNNSQSDVVSFRLSVEAQAQVTLNGVSKP
EAVLFPVSDWHPRDQPQKEEDLGPAVHHVYELINQGPSSISQGVLELSCPQALEGQQLLYV
TRVTGLNCTTNHPINPKGLELDPEGSLHHQQKREAPSRSSASSGPQILKCPEAECFRLRCE
LGPLHQQESQSLQLHFRVWAKTFLQREHQPFSLQCEAVYKALKMPYRILPRQLPQKERQVA
TAVQWTKAEGSYGLNDIFEAQKIEWHE

The yellow label indicates the SUMO tag.

Sequence Design：His-SUMO-Integrin β1-spycatcher AAseq：



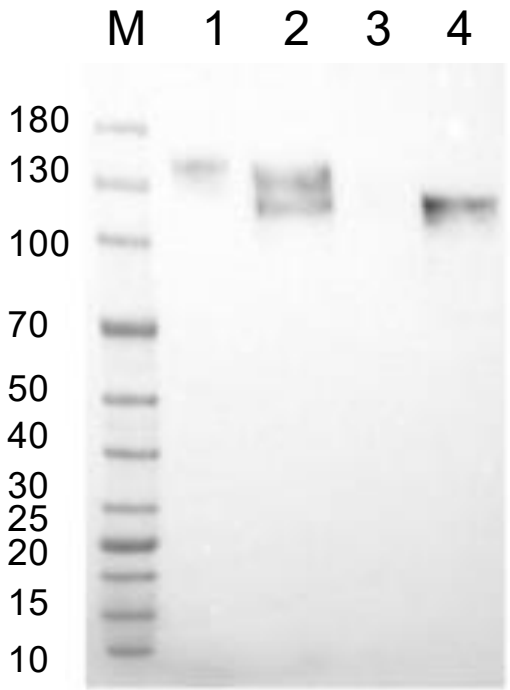
before enzymatic digestion
MW: 101.73 kDa
after enzymatic digestion
MW: 89.52 kDa

HHHHHHMSDSEVNQEAKPEVKPEVKPETHINLKVSDGSSEIFFKIKKTTPLRRLMEAFAK
RQGKEMDSLTFLYDGIEIQADQTPEDLDMEDNDIIEAHREQIGGGSQTDENRCLKANAKS
CGECIQAGPNCGWCTNSTFLQEGMPTSARCDDLEALKKKGCPPDDIENPRGSKDIKKN
KNVTNRSKGTAEKLKPEDITQIQPQQLVLRLRSGEPQTFTLKFKRAEDYPIDLYYLMDLSY
SMKDDLENVKSLGTDLMNEMRRITSDFRIGFGSFVEKTVMPYISTTPAKLRNPCTSEQNC
TSPFSYKNVLSLTNKGEVFNELVGKQRISGNLDSPEGGFDAIMQVAVCGSLIGWRNVTRL
LVFSTDAGFHFAGDGKLGGIVLPNDGQCHLENNMYTMSHYYDYPSIAHLVQKLSENNIQT
IFAVTEEFQPVYKELKNLIPKSAVGTLSANSSNVIQLIIDAYNSLSSEVILENGKLSEGVTISY
KSYCKNGVNGTGENGRKCSNISIGDEVQFEISITSNKCPKKDSDSFKIRPLGFTEEVEVIL
QYICECECQSEGIPESPKCHEGNGTFECGACRCNEGRVGRHCECSTDEVNSEDMDAY
CRKENSSEICSNNGECVCGQCVCRKRDNTNEIYSGKFCECDNFNCDRSNGLICGGNGV
CKCRVCECNPNYTGSACDCSLDTSTCEASNGQICNGRGICECGVCKCTDPKFQGQTCE
MCQTCLGVCAEHKECVQCRAFNKGEKKDTCTQECSYFNITKVESRDKLPQPVQPDPVS
HCKEKDVDDCWFYFTYSVNGNNEVMVHVVENPECPTGPDGGGGSGGGGSGGGGSD
SATHIKFSKRDEDGKELAGATMELRDSSGKTISTWISDGQVKDFYLYPGKYTFVETAAPD
GYEVATAITFTVNEQGQVTVNGKATKGDAHID*

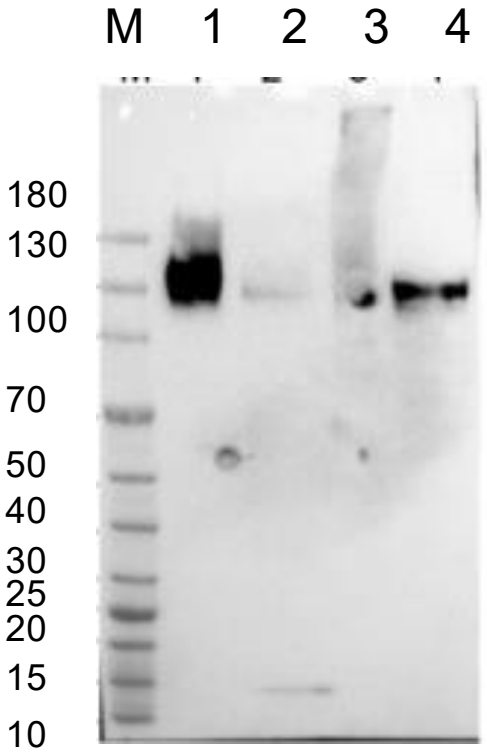
The yellow label indicates the His-SUMO tag.

results

1） The WB of Integrin α5β1 in the supernatant and cell.



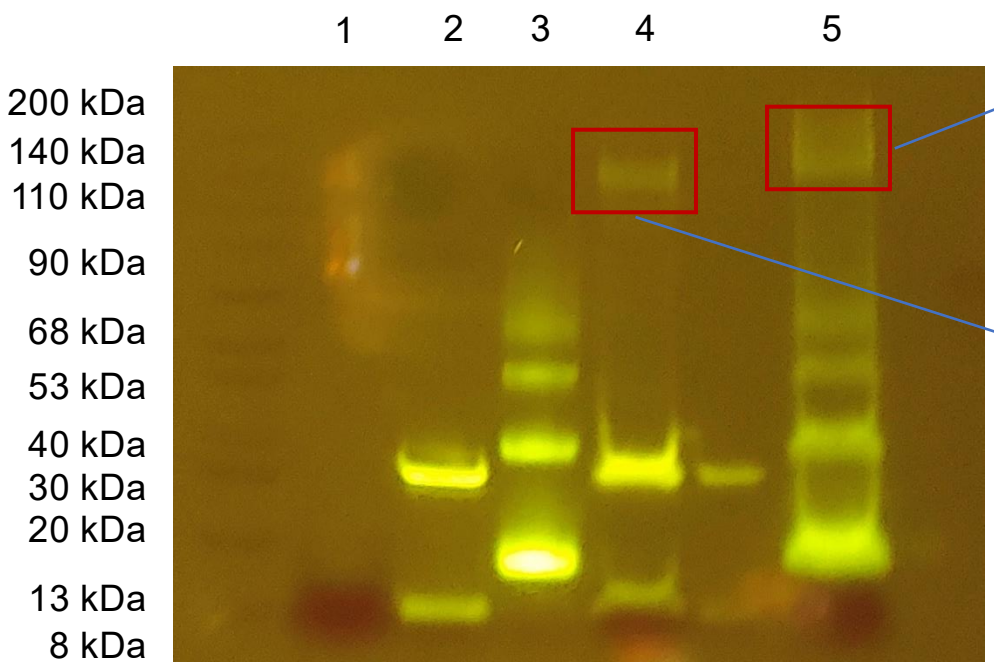
1 HM654-Biotin extracellular supernatant non-reduced 5ul
2 HM654-Biotin extracellular supernatant reduced 5ul
3 HM654-Biotin Intracellular supernatant non-reduced 5ul
4 HM654-Biotin Intracellular supernatant reduced 5ul
Antibody detection: His-HRP (1:50000) D250513C56



1 HM654-Biotin extracellular supernatant non-reduced 5ul
2 HM654-Biotin extracellular supernatant reduced 5ul
3 HM654-Biotin Intracellular supernatant non-reduced 5ul
4 HM654-Biotin Intracellular supernatant reduced 5ul
Antibody detection: SA-HRP (1:1000) D250513C56

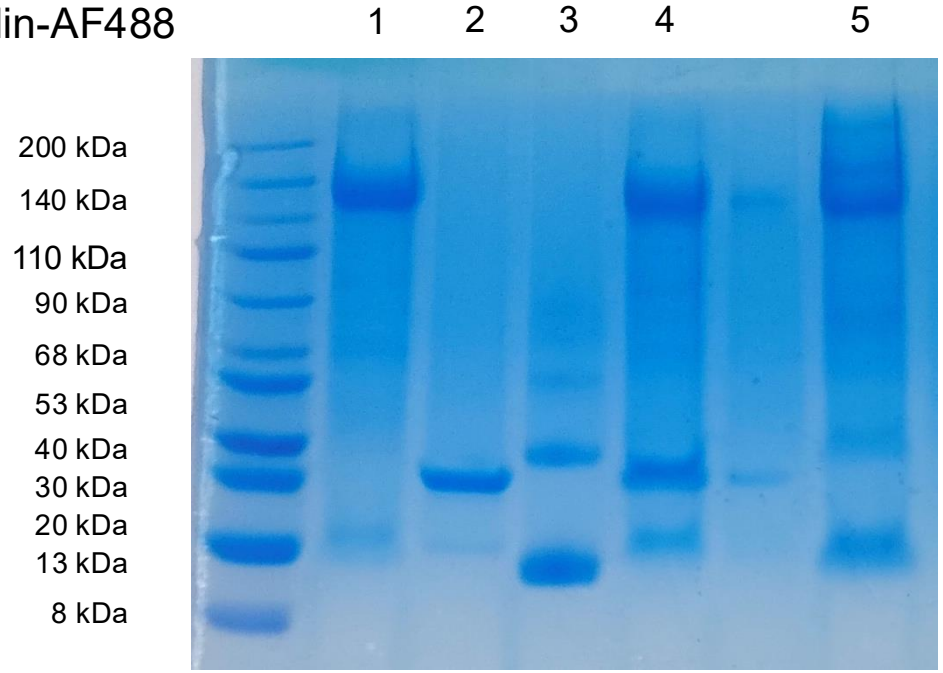
results

2) SDS-PAGE gel of integrin α5β1 Interacts with spytag-DNA-FAM and Streptavidin-AF488



Integrin α5-Streptavidin-AF488

Integrin β1-spytag-DNA-FAM



Lane	1	2	3	4	5
	Integrin α5β1	spytag-DNA-FAM	Streptavidin-AF488	Integrin α5β1 + spytag-DNA-FAM (1:10)	Integrin α5β1 + Streptavidin-AF488 (1:10)